A More Powerful Selective Inference Test for k-Means Clustering

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Abstract

After partitioning data into clusters with k-means, a natural next step is to test whether two estimated clusters have different means. Because the very hypothesis is chosen by looking at the data, a classical z- or χ^2-test is badly anti-conservative: on the Palmer penguins data its p-values for three cluster pairs are on the order of 10^{-57}, 10^{-29}, and 10^{-134}. Chen and Witten [2023] solve this by conditioning on the *entire* realized sequence of Lloyd-algorithm cluster assignments and computing an exact truncated χ p-value. Conditioning on the full algorithmic path is convenient but discards power. We make three contributions. First, following the "condition on less" principle of Chen et al. [2023] and the selective optimality theory of Fithian et al. [2017], we propose a test that conditions only on the event that the two *tested* clusters are preserved, taken as a union over all Lloyd paths that produce them; this enlarges the truncation set and is uniformly more powerful, the gain generally larger for more clusters K. Second, the same test extends to unknown noise variance: we studentize, replacing the χ statistic with a Fisher–F in which σ cancels, and characterize the conditioning set exactly for any K, generalizing Yun and Barber [2023] from $K = 2$ and eliminating their Monte Carlo step; the test requires no estimate of σ at all. Third, we give a fast numerical/analytic solver for the quadratic truncation inequalities that define the conditioning set [Yun and He, 2024]. In simulations the union test controls the selective Type I error (rejection rate near the nominal 0.05 at the global null versus 0.975 for the naive test) while substantially improving power, and a head-to-head comparison against data thinning [Neufeld et al., 2024] shows that our studentized test detects real clusters far more often — because data thinning must spend half the data on the test fold and must estimate σ to perform the split, whereas our test needs neither.

1 Introduction

Clustering is routinely used to generate hypotheses: one partitions the observations, notices that two groups look different, and then asks whether the difference is real. The trouble is that the groups, and hence the hypothesis, were chosen using the same data used to test it — a form of double dipping. A classical Wald test that ignores this selection has a wildly inflated Type I error rate [Gao et al., 2024, Chen and Witten, 2023]. The size of the problem is easy to see empirically: clustering the Palmer penguins data into $K = 3$ groups and applying a naive χ^2 test to the three cluster pairs gives p-values of 3.3×10^{-57}, 7.5×10^{-29}, and 5.6×10^{-134} (Section 7); these are absurd certificates of significance produced *purely* by the act of clustering.

Selective inference repairs this by conditioning on the selection event [Lee et al., 2016, Fithian et al., 2017]. For k-means clustering, Chen and Witten [2023] test $H_0 : \boldsymbol{\mu}^\top \boldsymbol{\nu} = \mathbf{0}$ for the contrast $\boldsymbol{\nu}$ encoding the difference in means of two estimated clusters, conditioning on the clustering output (and on nuisance directions) so that the conditional law of the test statistic is a truncated χ distribution. Their test controls the selective Type I error and is computable in closed form.

However, it conditions on the *full realized Lloyd path*: the entire sequence of intermediate cluster assignments produced by the algorithm, iteration by iteration. This is far more information than the hypothesis requires — the hypothesis only concerns the two clusters that were ultimately tested — and conditioning on it shrinks the truncation set and the power.

Contributions. This paper makes three contributions.

1. **A more powerful test that conditions on less** (Section 3). We condition only on the event that the two *tested* clusters are preserved, unioned over every Lloyd path that outputs them. This is the k-means analogue of the "condition on less $\Rightarrow$ more power" principle that Chen et al. [2023] used for the graph fused lasso, and is justified by the selective-optimality framework of Fithian et al. [2017]. The resulting truncation set is a superset of the path set, so the test is uniformly more powerful; empirically the power gain over the path test is generally larger for more clusters K.

2. **The same test when σ is unknown** (Section 4). Both the path and union tests extend to unknown variance with a single change — the statistic. Following Yun and Barber [2023] we studentize, replacing the χ statistic with an F in which σ cancels, and we characterize the conditioning set *exactly* for any K — by perturbing the data so that only the studentized statistic varies — replacing the importance sampling that Yun and Barber [2023] require for $K > 2$ with an exact, zero-variance p-value.

3. **A fast solver** (Section 5). The conditioning set is the solution set of a collection of quadratic inequalities in the perturbation parameter; we compute the coefficients analytically and solve by an interval/scan method, in the spirit of Yun and He [2024].

We also report a head-to-head comparison against data thinning [Neufeld et al., 2024, Dharamshi et al., 2024], a sample-splitting alternative to selective inference. Data thinning is valid, but it pays two prices that our studentized test avoids: it clusters on only half the data, so it detects fewer real clusters, and it needs an estimate of σ to perform the split — an estimate that is inflated two- to three-fold under genuine cluster separation, poisoning the split.

2 Background: the path test of Chen and Witten

2.1 Model and hypothesis

We observe $\mathbf{X} \in \mathbb{R}^{n\times q}$ with independent rows,

$$\mathbf{X} \sim \mathcal{MN}_{n\times q}(\boldsymbol{\mu}, \mathbf{I}_n, \sigma^2\mathbf{I}_q), \qquad \text{i.e.} \quad X_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \sigma^2\mathbf{I}_q), \tag{1}$$

with unknown mean rows $\boldsymbol{\mu}_i$ and (for now) known σ^2. Running k-means with K clusters returns a partition $\hat{\mathcal{C}}_1, \dots, \hat{\mathcal{C}}_K$ of $\{1, \dots, n\}$. Given two estimated clusters $\hat{\mathcal{C}}_1, \hat{\mathcal{C}}_2$ we test

$$H_0: \ \tfrac{1}{|\hat{\mathcal{C}}_1|}\sum_{i\in\hat{\mathcal{C}}_1}\boldsymbol{\mu}_i = \tfrac{1}{|\hat{\mathcal{C}}_2|}\sum_{i\in\hat{\mathcal{C}}_2}\boldsymbol{\mu}_i \quad\Longleftrightarrow\quad H_0: \ \boldsymbol{\mu}^\top\boldsymbol{\nu} = \mathbf{0}_q, \tag{2}$$

where the contrast $\boldsymbol{\nu} \in \mathbb{R}^n$ has entries $\nu_i = \mathbf{1}\{i \in \hat{\mathcal{C}}_1\}/|\hat{\mathcal{C}}_1| - \mathbf{1}\{i \in \hat{\mathcal{C}}_2\}/|\hat{\mathcal{C}}_2|$. The natural statistic is $\left\|\mathbf{X}^\top\boldsymbol{\nu}\right\|_2$, the length of the estimated difference in means.

2.2 The selective p-value and the Lloyd path

The naive Wald p-value $p_{\text{naive}} = \mathbb{P}(\left\|\mathbf{X}^\top \boldsymbol{\nu}\right\|_2 \geq \left\|\mathbf{x}^\top \boldsymbol{\nu}\right\|_2)$ ignores that $\boldsymbol{\nu}$ was chosen by clustering and is severely anti-conservative. Chen and Witten [2023] instead condition on the clustering output. Writing $c_i^{(t)}(\mathbf{X})$ for the cluster assigned to observation i at iteration t of Lloyd's algorithm, they define

$$p_{\text{path}} = \mathbb{P}_{H_0}\left(\left\|\mathbf{X}^\top \boldsymbol{\nu}\right\|_2 \geq \left\|\mathbf{x}^\top \boldsymbol{\nu}\right\|_2 \;\middle|\; \bigcap_{t=0}^{T}\bigcap_{i=1}^{n}\{c_i^{(t)}(\mathbf{X}) = c_i^{(t)}(\mathbf{x})\},\ \mathbf{P}_\nu^\perp \mathbf{X} = \mathbf{P}_\nu^\perp \mathbf{x},\ \operatorname{dir}(\mathbf{X}^\top \boldsymbol{\nu}) = \operatorname{dir}(\mathbf{x}^\top \boldsymbol{\nu})\right). \tag{3}$$

The conditioning event fixes everything except a single scalar: perturbing the data along $\boldsymbol{\nu}\operatorname{dir}(\mathbf{x}^\top\boldsymbol{\nu})^\top$, $\mathbf{x}'(\phi) = \mathbf{x} + \left((\phi - \left\|\mathbf{x}^\top \boldsymbol{\nu}\right\|_2)/\left\|\boldsymbol{\nu}\right\|_2^2\right)\boldsymbol{\nu}\operatorname{dir}(\mathbf{x}^\top\boldsymbol{\nu})^\top$, the p-value reduces to a one-dimensional truncated-χ probability,

$$p_{\text{path}} = \mathbb{P}\left(\phi \geq \left\|\mathbf{x}^\top \boldsymbol{\nu}\right\|_2 \mid \phi \in \mathcal{S}_T\right), \qquad \phi \sim (\sigma \left\|\boldsymbol{\nu}\right\|_2) \cdot \chi_q, \tag{4}$$

where $\mathcal{S}_T = \{\phi \geq 0 : \text{the entire Lloyd path is preserved at } \mathbf{x}'(\phi)\}$. The key computational fact is that $\mathcal{S}_T$ is the intersection of the solution sets of $O(nKT)$ quadratic inequalities in ϕ, so it is a union of intervals computable analytically; the test controls the selective Type I error in the sense of Lee et al. [2016] and Fithian et al. [2017].

2.3 The orthogonal decomposition

A geometric view, which we use throughout, decomposes the data with three Frobenius-orthogonal projections in $\mathbb{R}^{n\times n}$,

$$\mathbf{X} = \mathbf{P}_0\mathbf{X} + \mathbf{P}_1\mathbf{X} + \mathbf{P}_2\mathbf{X}, \tag{5}$$

where (i) the rank-one *between-cluster* projection $\mathbf{P}_0 = \boldsymbol{\nu}\boldsymbol{\nu}^\top / \left\|\boldsymbol{\nu}\right\|_2^2$ carries the contrast, with $\left\|\mathbf{P}_0\mathbf{X}\right\|_F = \left\|\mathbf{X}^\top\boldsymbol{\nu}\right\|_2 / \left\|\boldsymbol{\nu}\right\|_2$; (ii) the *within-cluster* centering projection $\mathbf{P}_1$ of the two tested clusters (rank $m-2$, with $m = |\hat{\mathcal{C}}_1| + |\hat{\mathcal{C}}_2|$) carries the within-cluster scatter; and (iii) the *nuisance* $\mathbf{P}_2 = \mathbf{I} - \mathbf{P}_0 - \mathbf{P}_1$. The path and union tests move only $\left\|\mathbf{P}_0\mathbf{X}\right\|_F$ along the contrast, holding $\mathbf{P}_1\mathbf{X}$ and $\mathbf{P}_2\mathbf{X}$ fixed; the studentized test of Section 4 instead varies the ratio of the two (Figure 1).

Why the path test loses power. Conditioning on the whole path is *sufficient* for validity but not *necessary*: the hypothesis depends only on the two tested clusters, not on the order in which Lloyd's algorithm reached them. Many distinct intermediate assignment sequences end in the same final partition of the two tested clusters; the path test discards all of them but the realized one. By Fithian et al. [2017], conditioning on this extra, ancillary information can only cost power.

3 A more powerful test: condition on less

3.1 The union conditioning set

Our test conditions on the *final-partition* event for the two tested clusters, rather than the full path. Define

$$\mathcal{S}_{\text{union}} = \left\{\phi \geq 0 :\ k\text{-means on } \mathbf{x}'(\phi) \text{ outputs clusters that preserve both } \hat{\mathcal{C}}_1 \text{ and } \hat{\mathcal{C}}_2\right\}. \tag{6}$$

Concretely, along the perturbation $\mathbf{x}'(\phi)$ we re-run k-means (with a shared initialization) and record, for each ϕ, whether the two tested clusters reappear — regardless of which Lloyd path produced them. Because the realized path is one of the paths that preserve the tested clusters,

$$\mathcal{S}_T \;\subseteq\; \mathcal{S}_{\text{union}}, \tag{7}$$

so $\mathcal{S}_{\text{union}}$ is a superset of the path set, and the corresponding truncated-χ p-value, $p_{\text{union}} = \mathbb{P}(\phi \geq \left\|\mathbf{x}^\top \boldsymbol{\nu}\right\|_2 \mid \phi \in \mathcal{S}_{\text{union}})$, is based on weaker conditioning. This is exactly the construction Chen et al. [2023] use for the graph fused lasso, transported to k-means: condition on the *tested object* (here, the two clusters) instead of the *algorithmic trace* (the Lloyd path).

3.2 Validity

Proposition 1. *Along the perturbation $\mathbf{x}'(\phi)$ the event $\{\mathbf{x}'(\phi)$ preserves both $\hat{\mathcal{C}}_1$ and $\hat{\mathcal{C}}_2\}$ is a deterministic function of ϕ, so $\mathcal{S}_{\text{union}}$ is a fixed (data-dependent) subset of $[0, \infty)$. Conditional on $\phi \in \mathcal{S}_{\text{union}}$ and on the nuisance coordinates, $\phi \sim (\sigma \|\boldsymbol{\nu}\|_2)\, \chi_q$ truncated to $\mathcal{S}_{\text{union}}$; hence p_{union} is super-uniform under H_0.*

The argument is exactly the one that validates the path test: any conditioning event that is a function of ϕ alone leaves the truncated-χ pivot intact (everything orthogonal to the contrast is held fixed), and $\mathcal{S}_{\text{union}} \supseteq \mathcal{S}_T$ only enlarges the truncation set, so the test stays valid while gaining power. Empirically the union test is calibrated at the global null (Section 6), against a rejection rate of 0.975 for the naive test.

3.3 Construction of the union set

We enumerate the distinct Lloyd paths reachable along the perturbation line as follows. Starting from the observed $\phi = \left\|\mathbf{x}^\top \boldsymbol{\nu}\right\|_2$, we compute the ϕ-interval on which the observed path is realized (the path test's $\mathcal{S}_T$), then probe just past each interval endpoint, re-run k-means there to discover the neighbouring path, compute *its* preservation interval, and repeat, queueing newly discovered endpoints. The union of all intervals whose path preserves both tested clusters is $\mathcal{S}_{\text{union}}$; the subset of intervals matching the observed path signature recovers $\mathcal{S}_T$ exactly, so the two p-values are produced from a single sweep. The exploration is purely Euclidean (it perturbs $\mathbf{x}$ and re-runs Euclidean k-means), which makes it *metric-independent*: the known general-covariance test $\mathbf{X} \sim \mathcal{MN}(\boldsymbol{\mu}, \mathbf{I}_n, \boldsymbol{\Sigma})$ reuses the identical exploration and only rescales the resulting intervals by a scalar and swaps the test statistic for $\sqrt{(\mathbf{x}^\top \boldsymbol{\nu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}^\top \boldsymbol{\nu})}$. The general-covariance path p-value matches the isotropic one to machine precision.

A scan cutoff. The far tail of the perturbation line contributes negligible χ mass to the p-value. We therefore stop exploring once ϕ exceeds the point where the conditional χ survival mass beyond it falls below a fraction `scan_tail_prob` (default 10^{-6}) of the survival mass beyond the observed statistic. This leaves the p-value essentially unchanged (verified $\Delta \leq 10^{-11}$) while cutting roughly 25–40% of the path exploration.

4 Unknown variance: the same test, studentized

4.1 The F pivot

Nothing about the conditioning changes when σ is unknown — we still test the path or the union of preserved-cluster events — only the statistic does. When σ is unknown we cannot compare $\left\|\mathbf{x}^\top \boldsymbol{\nu}\right\|_2$ to a scaled χ_q, because the scale is exactly what we lack. Following Yun and Barber [2023] we studentize. Using the decomposition of Section 2.3, the within-cluster scatter $\|\mathbf{P}_1 \mathbf{X}\|_F^2$ is an unbiased σ^2-estimate independent of the between-cluster signal $\|\mathbf{P}_0\mathbf{X}\|_F^2$, so their ratio is pivotal:

$$R \;=\; \frac{\|\mathbf{P}_0\mathbf{X}\|_F^2\,/q}{\|\mathbf{P}_1\mathbf{X}\|_F^2\,/\big((m-2)q\big)} \;=\; (m-2)\,\frac{\|\mathbf{P}_0\mathbf{X}\|_F^2}{\|\mathbf{P}_1\mathbf{X}\|_F^2} \;\sim\; F_{q,\,(m-2)q} \quad \text{under } H_0', \tag{8}$$

where $m = |\hat{\mathcal{C}}_1| + |\hat{\mathcal{C}}_2|$ counts only the two tested clusters and the unknown σ cancels. The required null is the slightly stronger *pointwise* null H_0': within each tested cluster the means $\boldsymbol{\mu}_i$ are equal to the cluster mean (so that $\|\mathbf{P}_1\mathbf{X}\|_F^2$ is a clean noise estimate). This is the price of not knowing σ. Relative to H_0 (equal cluster means), H_0' additionally asks for within-cluster mean homogeneity. A violation that still satisfies H_0 — for instance canceling $\pm$ sub-populations with equal cluster means — only *inflates* the denominator $\|\mathbf{P}_1\mathbf{X}\|_F^2$, making the test *conservative*; anti-conservativity would instead require within-cluster heterogeneity that *deflates* $\|\mathbf{P}_1\mathbf{X}\|_F^2$ under a true mean difference, a configuration k-means tends to resolve by spending a cluster. We therefore treat H_0' as a benign strengthening of H_0 in practice.

4.2 Conditioning when σ is unknown

The conditioning set is built the same way as before — we ask which perturbations of the data preserve the tested clusters — but the perturbation must now hold fixed everything the F statistic treats as nuisance and vary *only* the ratio R. Concretely we condition on the clustering, on $\mathbf{P}_2\mathbf{X}$, on the unit directions $U_0 = \mathbf{P}_0\mathbf{X}/\,\|\mathbf{P}_0\mathbf{X}\|_F$ and $U_1 = \mathbf{P}_1\mathbf{X}/\,\|\mathbf{P}_1\mathbf{X}\|_F$ (which are orthonormal), and on the *total sum of squares* $T = \|\mathbf{P}_0\mathbf{X}\|_F^2 + \|\mathbf{P}_1\mathbf{X}\|_F^2$. The single remaining degree of freedom is an angle θ, and the conditional perturbation is

$$\mathbf{x}(\theta) \;=\; \mathbf{P}_2\mathbf{x} \;+\; \sqrt{T}\,\big(\sin\theta\, U_0 + \cos\theta\, U_1\big), \qquad \mathbf{x}(\theta_{\text{obs}}) = \mathbf{x}, \tag{9}$$

with $\theta_{\text{obs}} = \operatorname{atan2}(\|\mathbf{P}_0\mathbf{x}\|_F\,, \|\mathbf{P}_1\mathbf{x}\|_F)$ and $R(\theta) = (m-2)\tan^2\theta$. Because $U_0 \perp U_1$ are orthonormal, (9) is a **Givens rotation** of the vectorized data $\mathbf{x} \in \mathbb{R}^{nq}$ inside the plane $\operatorname{span}\{U_0, U_1\}$, fixing its orthogonal complement,

$$\mathbf{x}(\theta) = G(\theta - \theta_{\text{obs}})\,\mathbf{x}, \qquad G(\delta) = \mathbf{I} + (\cos\delta - 1)\big(U_0U_0^\top + U_1U_1^\top\big) + \sin\delta\,\big(U_0U_1^\top - U_1U_0^\top\big). \tag{10}$$

Since G is orthogonal it preserves the in-plane radius, so T is held fixed and only the ratio R varies — which is precisely why the test is free of σ: under H_0', T is complete and sufficient for σ while R is scale-ancillary, so Basu's theorem gives $R \perp T$ and conditioning on T removes σ. This is the *only* change from the known-σ tests, which instead move along the contrast — varying the separation $\|\mathbf{P}_0\mathbf{X}\|_F$ while the within-cluster scatter $\mathbf{P}_1\mathbf{X}$ stays put. Figure 1 draws the two: the known-σ tests vary the separation ($\to \chi$), the studentized test varies the ratio at fixed total T ($\to F$). Geometrically, fixing T confines the data to a sphere of radius $\sqrt{T}$ on which the studentized perturbation rotates at fixed radius, so its statistic depends only on the rotation angle θ ($R = (m-2)\tan^2\theta$).

4.3 Path and union, studentized

With $\mathcal{S} = \{\theta \in (0, \pi/2) : \text{the clustering is preserved at } \mathbf{x}(\theta)\}$, the selective p-value is the truncated-F probability

$$p = \mathbb{P}\big(R(\theta) \geq R \mid \theta \in \mathcal{S}\big), \qquad R \sim F_{q,(m-2)q}. \tag{11}$$

The cluster-assignment margins $\|x_i - m_j\|^2 - \|x_i - m_k\|^2$ are quadratic in $(\sin\theta, \cos\theta)$, so $\mathcal{S}$ is a union of intervals of θ computed exactly (Section 5). Conditioning on the observed interval alone gives the studentized *path* test; taking the union of the intervals across all Lloyd paths that preserve the two tested clusters gives the studentized *union* test — exactly the path-versus-union choice of Section 3, now in the coordinate θ. Both are formed in that single coordinate, so the truncated-F stays a valid pivot.

The union must be formed in the studentized statistic's own coordinate. A tempting shortcut is to reuse the known-σ conditioning set — rescaled — inside the F pivot. This is *invalid*. That set was traced by perturbations along the contrast, which hold the within-cluster scatter $\|\mathbf{P}_1\mathbf{X}\|_F$ fixed; but the studentized F uses $\|\mathbf{P}_1\mathbf{X}\|_F^2$ to estimate σ, and the per-cluster cancellation that makes R pivotal relies on that scatter being free to vary with R. The F is therefore not a valid pivot over the imported set: the per-path truncated-F's aggregate into a union-F only when the union is taken among the perturbations (9) that hold T fixed, where $\{\text{clusters preserved}\} = \{R \in \mathcal{S}'\}$ is a function of R alone and the union of a one-dimensional truncation is exact. Exhaustive simulation confirms the rescaled import over-rejects (rejection 0.081, growing with the set's width from 0.010 up to 0.170), whereas the union formed correctly is calibrated.

Generalizing Yun and Barber to any K. For $K = 2$ the two tested clusters are all the data ($m = n$) and Yun and Barber [2023] compute the truncation set exactly by reusing Chen and Witten [2023]. For $K > 2$ they resort to importance sampling. Our exact solver computes the conditioning set exactly for any K, so it (i) replaces their Monte Carlo with an exact, zero-variance p-value for $K > 2$ k-means (even for the path test), and (ii) yields the more powerful union for free. The method is k-means-specific (theirs applies to general clustering, e.g. hierarchical) and requires the pointwise null H_0'.

When does the union help under unknown variance? The union's power gain is largely a *known-variance* phenomenon. A one-sided truncated p-value shrinks when enlarging the truncation set only if the added mass lands *below* the observed statistic. For the known-σ χ statistic, alternative Lloyd paths tolerate smaller separation and so extend the conditioning interval downward, below the observed value — producing large gains. For the studentized F, k-means has *minimized* the within-scatter $\|\mathbf{P}_1\mathbf{X}\|_F$, pinning R_{obs} against its low-R floor, so the extra room is almost entirely *above* the observed statistic and the union p-value $\approx$ the path p-value. In short, the union's leverage is *scale* (contrast-length) information, which a scale-free statistic discards. This is why the union gain shows up primarily under known σ (Figure 3) while the studentized test earns its place as the σ-free route to validity and to the Yun–Barber power regime (Figure 4).

5 Computation

Both conditioning sets reduce to solving quadratic inequalities in a single perturbation parameter.

Known σ. Along $\mathbf{x}'(\phi)$ each k-means assignment margin is a quadratic in ϕ, so the per-path preservation set is the intersection of $O(nKT)$ quadratic inequalities, computed in closed form, and the union set is assembled by the endpoint-probing sweep of Section 3.

Unknown σ. Along $\mathbf{x}(\theta)$ each assignment margin is an exact quadratic form $Q(a, b) = Aa^2 + Bb^2 + Cab + Da + Eb + F_c$ in $(a, b) = (\sin\theta, \cos\theta)$. Substituting $t = \tan(\theta/2)$ turns $Q(\sin\theta, \cos\theta)(1 + t^2)^2$ into a *quartic* $N(t)$, whose real roots in $[0, 1]$ give the breakpoints in θ; the sign of Q at a midpoint decides which segments satisfy the inequality, and intersecting over all margins yields the preserved set for one Lloyd trajectory (`sims/rfiber_exact.R`). The truncated-F mass over the resulting R-intervals is computed in a cancellation-proof way (taking the maximum of the CDF-difference and the survival-difference, the latter stable in the extreme upper tail under strong separation). The solver is modular and unit-tested: 23/23 tests for the F-region routine (including agreement with Monte Carlo and extreme-tail underflow), 14/14 for the seven primitive geometric modules, and 18/18 for the trajectory assembler; the truncated-F pivot is verified uniform under fixed truncation. A pure interval/scan implementation also computes the set without analytic quartics, agreeing with the exact solver where the grid is fine and finding more segments where the grid is coarse. The same numerical-solver idea underlies the multiple-pairs extension of Yun and He [2024].

6 Simulations

Unless noted, data are q-dimensional Gaussian with isotropic unit variance; k-means uses a fixed shared initialization. We report rejection rates at level 0.05.

6.1 Type I error

At the global null ($\boldsymbol{\mu} = \mathbf{0}$) every selective test is calibrated while the naive test is invalid (Figure 2): the selective p-values track the 45° line, the naive p-values collapse below it. Across $q \in \{2, 10, 50, 100\}$ the known-σ path/union reject at 0.032–0.063 and the studentized test at 0.034–0.058 (slightly conservative at $q = 50$), against a naive rejection rate of 0.99–1.00. Under AR(0.5)/AR(0.9) covariance the general-covariance union and path remain on the diagonal while the naive test flatlines (Figure 8); under heavy-tailed t_5/t_{10} noise the studentized test is fairly robust, with mild anti-conservatism only for the heaviest tails (Figure 9).

6.2 Power: union versus path

Figure 3 plots power against the separation δ, faceted by q. The known-σ union (dashed) is uniformly more powerful than the path (solid), with gains of order 0.1–0.4 (e.g. at $q = 2$, $\delta = 4$ the union rejects 0.795 versus the path's 0.400); averaged over $\delta > 0$ the union improves on the path by about 16.6 percentage points. The gain is *generally larger for larger K*: at $\delta = 5$, $q = 2$ the union's advantage over the path is 0.075/0.115/0.225/0.200 for $K = 2/3/4/5$ (`sims/results/sweep_standard`), a distinctive feature of our exact solver, which handles any K. Following Chen and Witten [2023] we separate *detection probability* (whether k-means recovers the two true clusters) from *conditional power* (rejection given detection); the union gain persists conditionally, confirming it is a property of the test, not the clustering (Figure 7). As anticipated in Section 4, the studentized union and path nearly coincide — the union's leverage is studentized away — so under unknown σ the contribution is the validity and power of the studentized test itself, not a union-over-path improvement.

6.3 Unknown variance: reproducing the Yun–Barber regime

We reproduce the equilateral-triangle, $K = 3$, $q \in \{2, 10\}$ design of Yun and Barber [2023] in our k-means framework (path test, clusters 1 vs 2, 300 reps; `repro_yunbarber_q{2,10}.rds`), comparing four variance handlings on the *same* truncation set: an oracle (known σ), the studentized F, a median-based $\hat{\sigma}_{\mathrm{MED}}$ plug-in, and a sample-SD plug-in. The ordering oracle $\geq$ studentized $\geq$ plug-ins holds throughout. The studentized F decisively beats the sample plug-in once the clusters separate: at $q = 2$, $\delta = 7$, power is 0.490 (studentized) versus 0.290 (sample) and 0.717 (oracle); the gap opens at $\delta \approx 4$–5. The mechanism is that the global sample SD absorbs the between-cluster separation, so $\mathbb{E}[\hat{\sigma}_{\text{sample}}]$ climbs roughly linearly in δ and the plug-in power plateaus, whereas the within-cluster scale stays near 1 and the F approaches the oracle. The robust $\hat{\sigma}_{\text{MED}}$ is dimension-dependent: near-oracle at $q = 10$ but, with three balanced clusters, as inflated as the sample SD at $q = 2$.

6.4 Comparison with data thinning

Data thinning [Neufeld et al., 2024] is a sample-splitting alternative: split each entry $X = X^{(1)} + X^{(2)}$ into independent Gaussian halves, cluster on $X^{(1)}$, and test on $X^{(2)}$ with an ordinary (unconditional) mean test. It is valid by construction, but it pays two prices our studentized test avoids. *First*, it clusters on only half the information, so it recovers the true pair less often. *Second*, to perform the split it needs a variance: the oracle (σ^2) is unavailable in practice, and any plug-in is inflated under cluster separation, which poisons both folds.

Figures 10 and 11 report the head-to-head, per-replicate matched, on a $K = 3$ equilateral-triangle design ($q = 2$, $n_{\text{per}} = 10$, $\varepsilon = 0.5$; `datathin_compare.rds`, null 400 reps, signal 220 reps each). Three things stand out. (i) At the global null all selective tests and oracle thinning control Type I; the plug-in thinning variants are slightly anti-conservative (0.075 for the MED split). (ii) *Detection probability* — whether the clustering recovers the true pair — is far higher for the full-data selective pipeline (e.g. 0.91 at $\delta = 4$) than for plug-in thinning, which clusters on half the data *and* with an inflated split variance, collapsing detection to 0.08–0.23. (iii) Even *conditional* on detection, the selective union and studentized tests are competitive with or better than plug-in thinning, while needing no σ. The recorded variance estimates make the poisoning explicit: under separation the MED and sample estimates inflate to 2.08–3.12 (true $\sigma = 1$) as δ runs $4 \to 7$. Only oracle thinning — which knows the true σ and so is not realizable — keeps high detection. Our studentized test needs no σ at all and clusters on the full data. Figure 10 draws all three metrics in the paper's shared grammar (colour = variance handling; line style = paradigm, selective solid vs. thinning dashed): panel A shows the Type I calibration as a QQ plot against the uniform, where every valid method — selective *and* thinning — tracks the 45° line while the naive test plunges below it, and panels B–C plot detection probability and conditional power against δ.

7 Real data

7.1 Palmer penguins

We cluster the Palmer penguins data [Horst et al., 2020] — four standardized morphological features, the same data used by Yun and Barber [2023] — into $K = 3$ groups and test all three cluster pairs (`penguins_pvalues_k3.csv`, $\hat{\sigma}_{\text{MED}} = 1.155$). Table 1 shows the pattern. The naive test reports absurd p-values (3.3×10^{-57}, 7.5×10^{-29}, 5.6×10^{-134}). The original known-σ *path* test, with the MED plug-in, fails to reject *any* pair (0.92, 0.51, 0.81) — conditioning too restrictively, and with an inflated $\hat{\sigma}$. The known-σ *union* recovers the clearest split (2 vs 3: $0.81 \to 1.6 \times 10^{-14}$) while staying cautious on the closer pairs (0.063, 0.27). The studentized test, needing no σ, rejects all three (2.3×10^{-5}, 0.028, 1.4×10^{-5}).

7.2 Single-cell RNA sequencing (Pollen)

We cluster 216 developing-cortex cells from Pollen et al. [2014] (balanced across four inferred cell types; CP10k normalization, log1p, top-500 highly variable genes, ten standardized PCs) into $K = 3$ groups and test the three pairs (`scrna_pvalues_k3.csv`). Here the union is the hero: the original

Table 1: Palmer penguins, $K = 3$, p-values for the three cluster pairs (`penguins_pvalues_k3.csv`). The naive test is absurd; the path test rejects none; the union recovers pair 2 vs 3; the studentized test rejects all three. Figure 5.

pair	naive	path (known)	union (known)	studentized F
1 vs 2	3.3×10^{-57}	0.920	0.063	2.3×10^{-5}
1 vs 3	7.5×10^{-29}	0.514	0.271	0.028
2 vs 3	5.6×10^{-134}	0.808	1.6×10^{-14}	1.4×10^{-5}

Table 2: Pollen scRNA-seq, $K = 3$, p-values for the three cluster pairs (`scrna_pvalues_k3.csv`). The path test rejects none; the union rejects all three. Figure 6.

pair	naive	path (known)	union (known)	studentized F
1 vs 2	4.6×10^{-34}	0.475	0.0061	0.024
1 vs 3	3.5×10^{-31}	0.179	0.041	1.000
2 vs 3	1.6×10^{-27}	0.067	0.043	1.000

path test rejects *none* of the three pairs (0.475, 0.179, 0.067), whereas the *union* rejects *all three* (0.0061, 0.041, 0.043). The studentized test rejects the clearest pair (1 vs 2: 0.024), and the naive p-values are again absurd (10^{-34}, 10^{-31}, 10^{-27}). Penguins and scRNA are complementary: on penguins the studentized test rejects the most pairs, on scRNA the known-σ union does — but in both cases the original path test is too conservative to be useful.

8 Discussion

We have given a more powerful selective test for k-means clustering that conditions on the two tested clusters rather than the full Lloyd path, an exact studentized version that needs no estimate of σ and extends Yun and Barber [2023] to any K, and a fast solver for both. The σ-free property is the practical headline: the most common failure of plug-in approaches — including data thinning's split step — is that a global variance estimate is inflated by exactly the cluster separation one is trying to detect, and the studentized test sidesteps this entirely.

Several limitations remain. The studentized test requires the pointwise null H_0' (equal means *within* each tested cluster), stronger than the equal- cluster-means null; this is the inherent cost of not knowing σ, and a violation is conservative unless within-cluster heterogeneity deflates the residual under a true mean difference. The union's power gain is a known-σ phenomenon and is "studentized away" under the F pivot, because k-means minimizes the within-scatter and pins the statistic against its floor. The method is specific to k-means / Lloyd's algorithm, whereas Yun and Barber [2023] and Gao et al. [2024] cover broader clustering families. Natural next steps are control of the family-wise error across all cluster pairs, building on Yun and He [2024], and randomized or data-fission variants that trade a little of the conditioning for extra power.

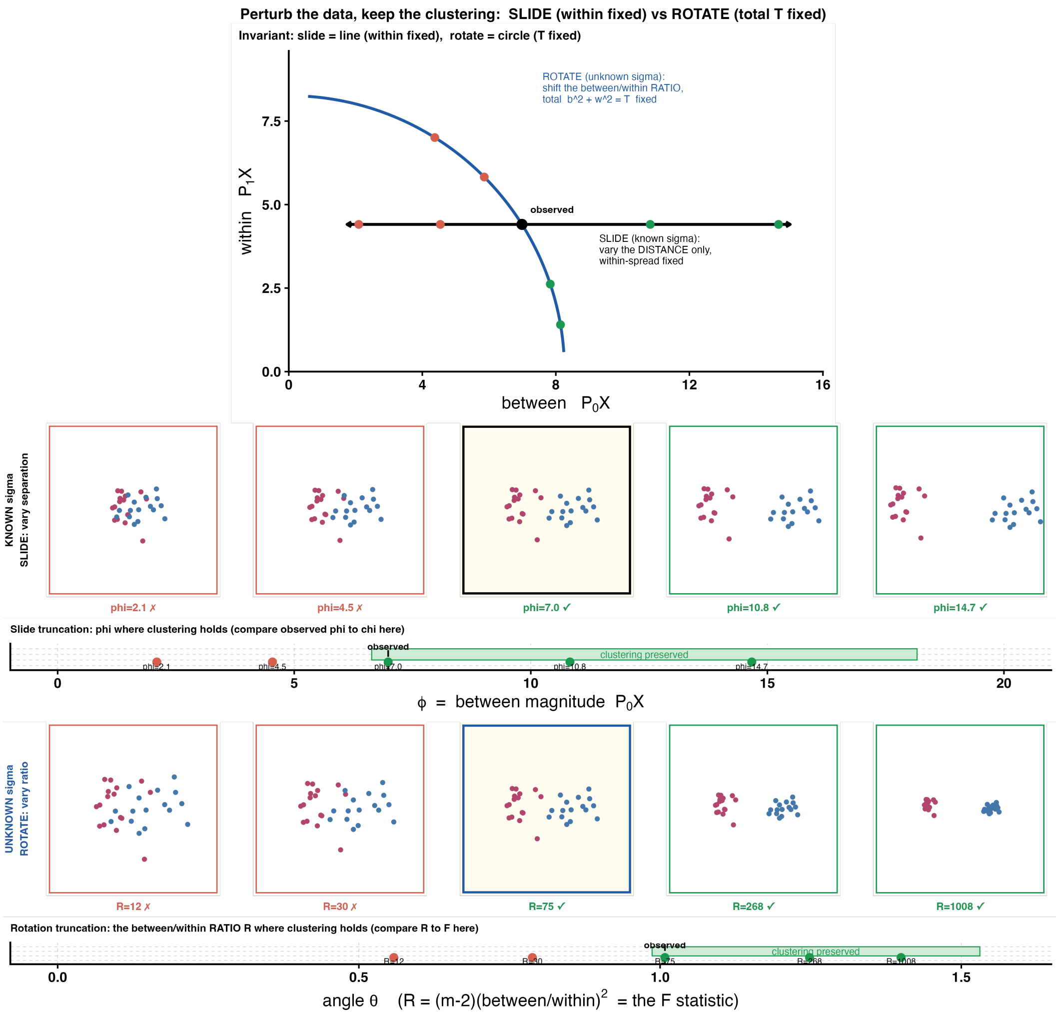

Figure 1: The contrast plane ($\|\mathbf{P}_0\mathbf{X}\|_F$, $\|\mathbf{P}_1\mathbf{X}\|_F$). The known-$\sigma$ tests fix the within-cluster scatter and vary the separation ($\to \chi$); the studentized test fixes the total T and varies the ratio R ($\to F$). Source: `sims/plot_decomposition.R` $\to$ `decomposition.png`.

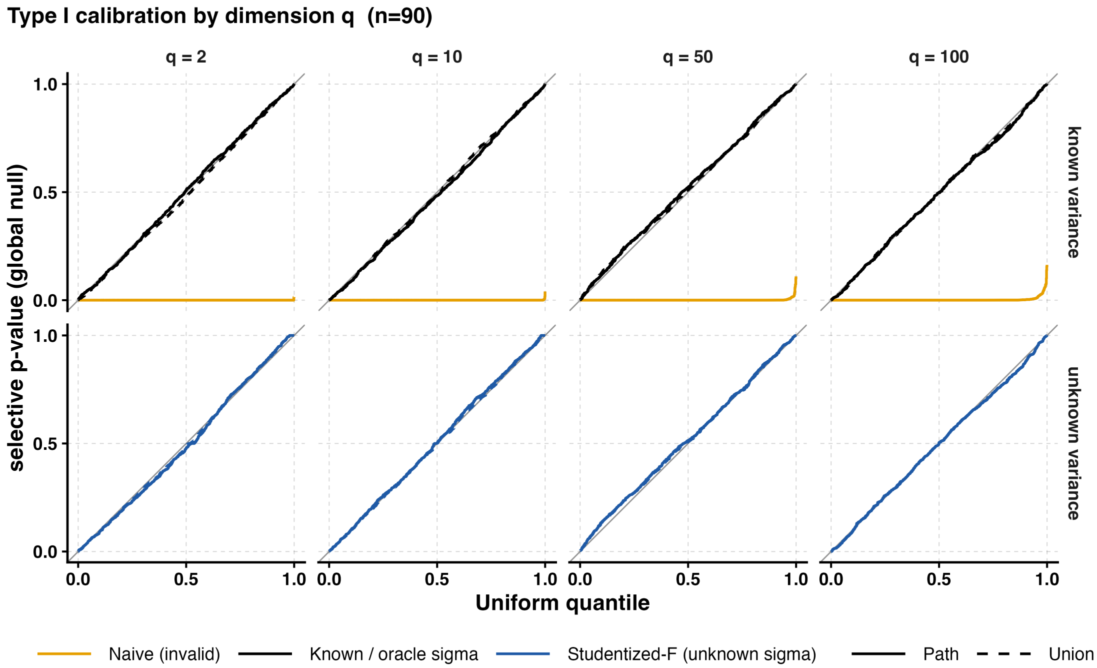

Figure 2: Selective Type I error ($n = 90$, $K = 3$). QQ-plots of the null p-values against Uniform$(0, 1)$, faceted by dimension q. **Top:** known-σ tests; **bottom:** studentized tests. The selective path (solid) and union (dashed) lie on the 45° line at every q, while the naive p-value (orange) collapses below it. Source: `sims/plot_cw_typeI.R` → `cw_typeI.png`.

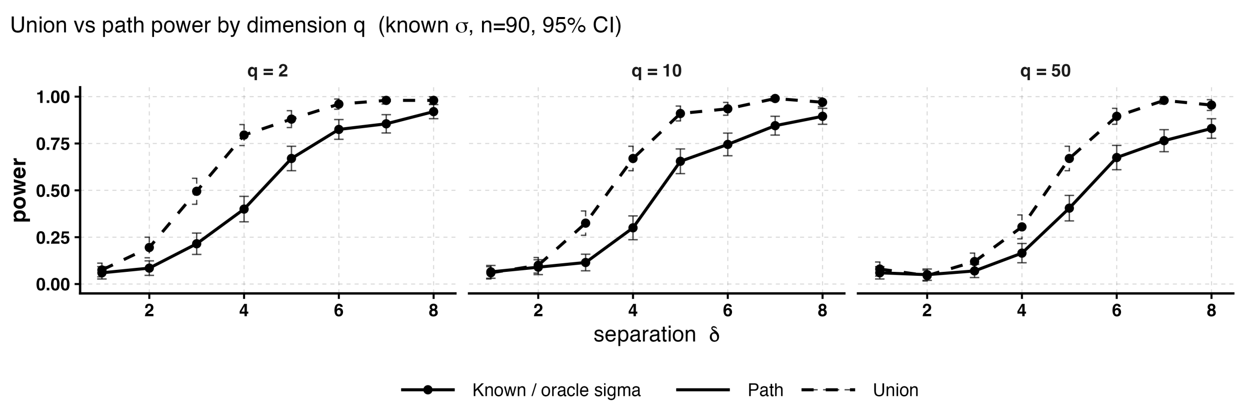

Figure 3: Power against separation δ (known σ, $n = 90$, $K = 3$, 95% CI), faceted by q. The union (dashed) is uniformly more powerful than the path (solid); the gap is largest in the moderate-signal regime. Source: `sims/plot_cw_power.R` → `cw_power.png`.

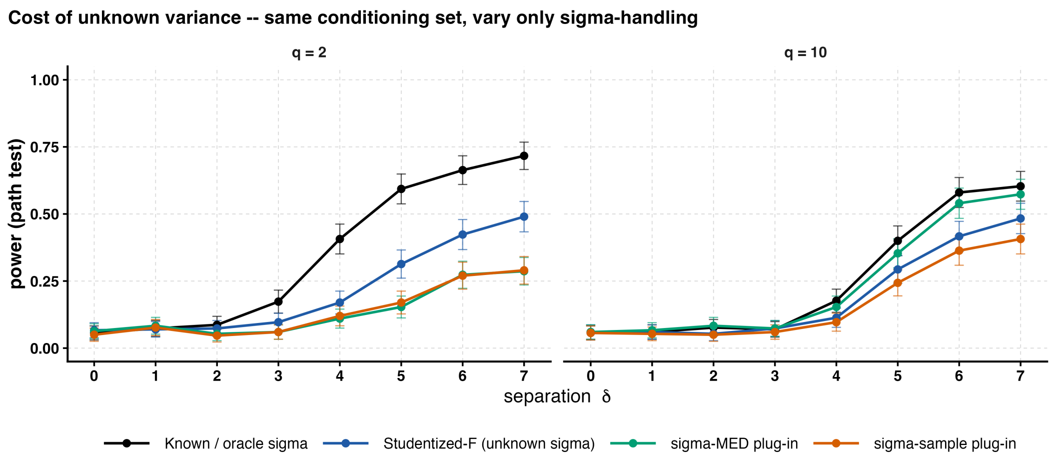

Figure 4: Reproducing the Yun–Barber regime on the same truncation set: oracle $\geq$ studentized F $\geq$ plug-ins. The studentized F beats the sample plug-in once clusters separate, because the global SD absorbs the separation. Source: `repro_yunbarber.png`.

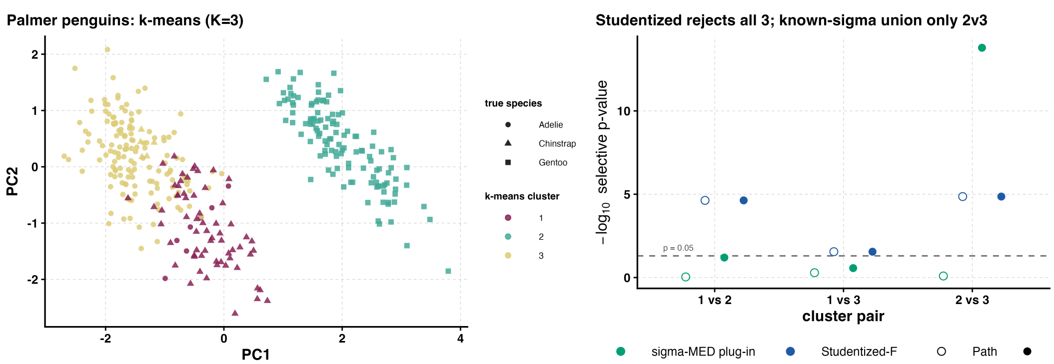

Figure 5: Palmer penguins, $K = 3$ (Table 1). Source: `penguins.png`.

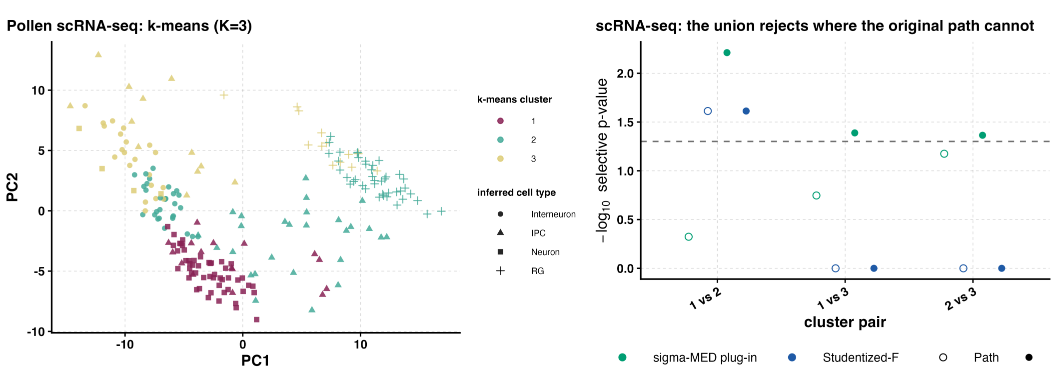

Figure 6: Pollen scRNA-seq, $K = 3$ (Table 2): the union rejects all three pairs, the path rejects none. Source: `scrna_k3.png`.

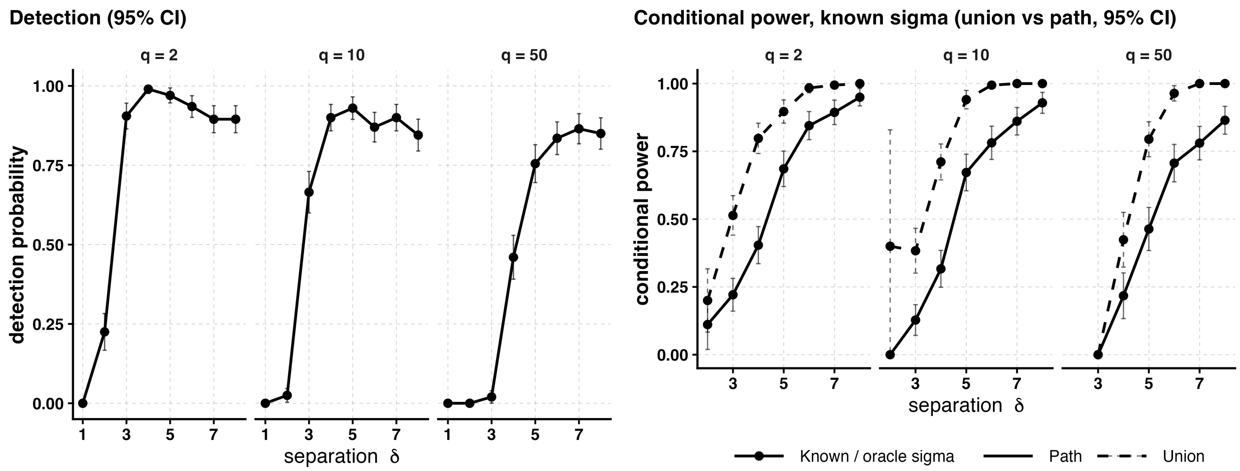

Figure 7: (Supplementary) Decomposition of power into **detection probability** (left; the chance k-means recovers the two true clusters, a property of the clustering) and **conditional power** (right; rejection given detection, a property of the test), against separation δ, faceted by q. The union (dashed) beats the path (solid) *conditionally*, so the gain in Figure 3 is attributable to the test, not the clustering. Source: `sims/plot_cw_power.R` → `cw_detection.png`.

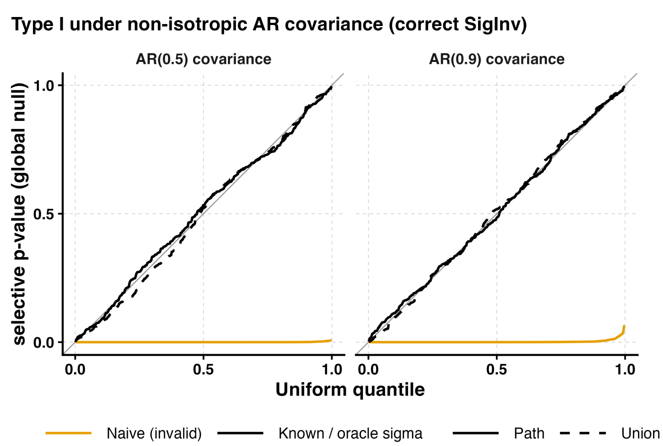

Figure 8: (Supplementary) Type I under general covariance. Under AR(0.5)/AR(0.9) covariance the general-covariance union/path stay on the 45° line while the naive test is invalid (the isotropic Type I is Figure 2). Source: `gencov_typeI.png`.

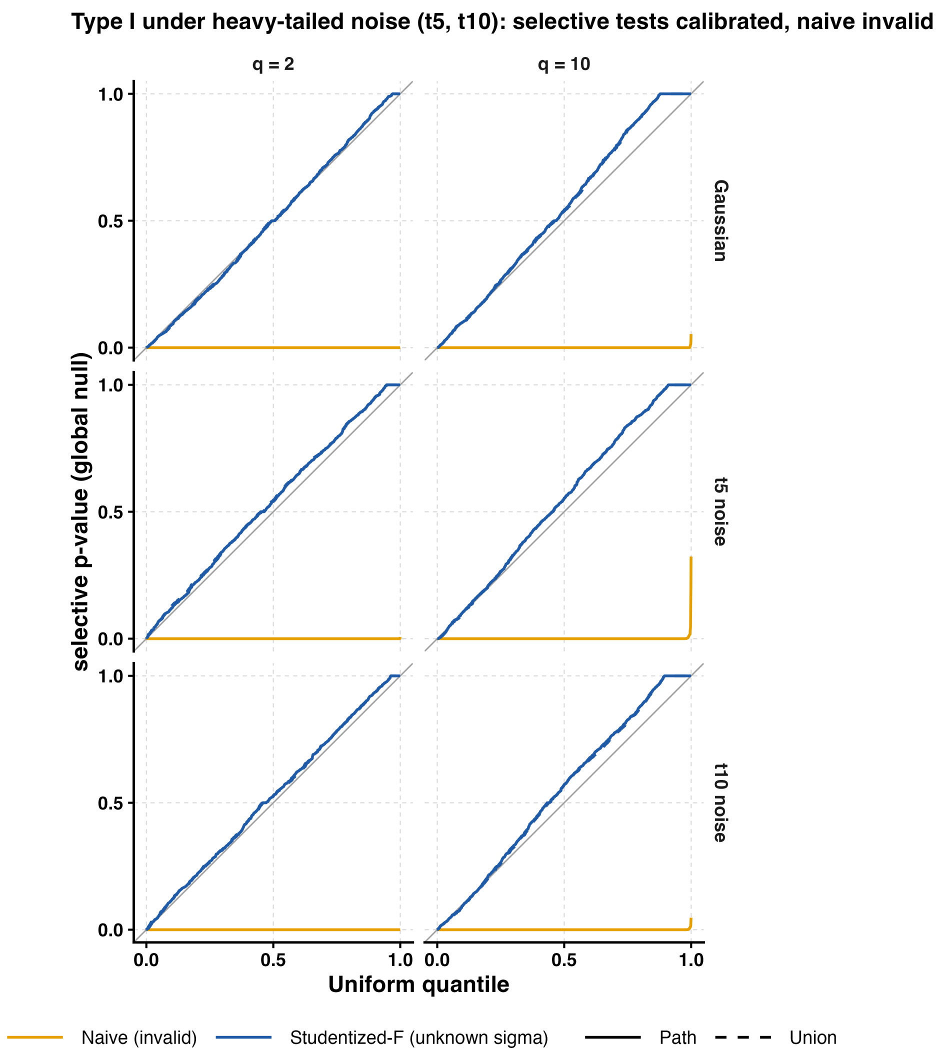

Figure 9: (Supplementary) Type I under heavy-tailed t_5/t_{10} noise. The studentized test is fairly robust; mild anti-conservatism only for the heaviest tails. Source: `heavytail_typeI.png`.

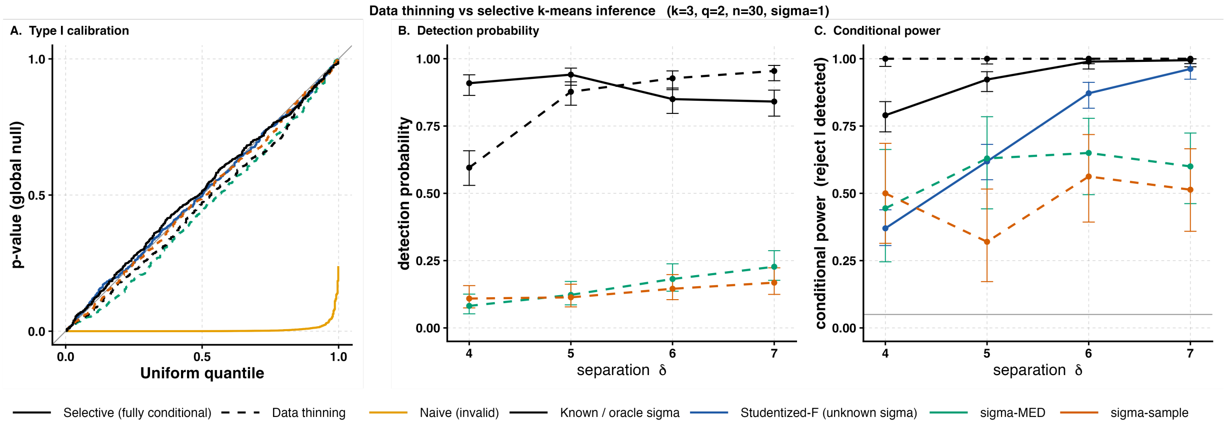

Figure 10: (Supplementary) Data thinning vs. the selective tests, all three metrics in the paper's shared grammar (colour = variance handling; line style = paradigm, selective solid vs. data thinning dashed). **A:** Type I as a QQ plot — every valid method tracks the 45° line, the naive test (orange) plunges below. **B:** detection probability vs. δ — full-data selective clustering (solid black) dominates the thinned splits, and the plug-in splits ($\hat{\sigma}_{\mathrm{MED}}$ green, $\hat{\sigma}_{\mathrm{sample}}$ vermilion) collapse because the variance estimate is inflated by the very separation being tested. **C:** conditional power vs. δ (reject | detected); oracle thinning rejects whenever it detects, but rarely detects. Source: `sims/plot_datathin.R` → `datathin.png`.

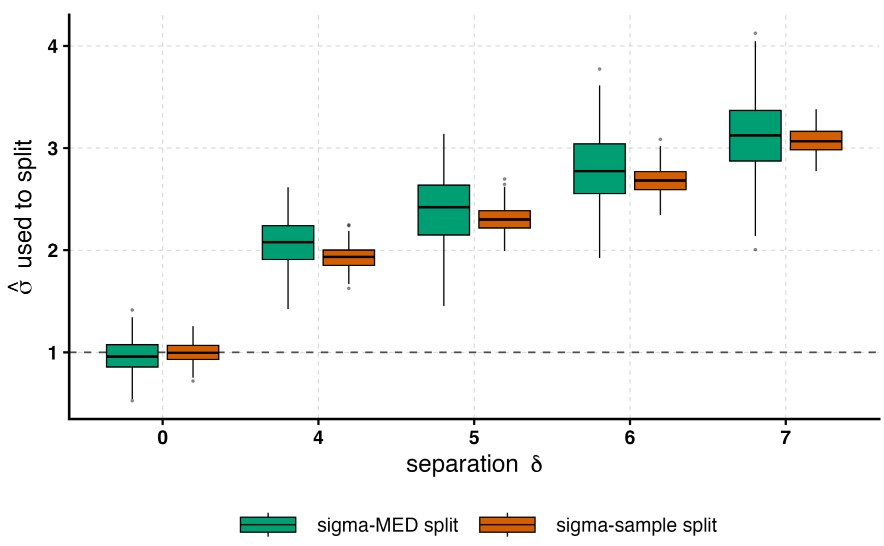

Figure 11: (Supplementary) Why data thinning's split is poisoned (`datathin_compare.rds`, $K = 3$, $q = 2$, $n_{\mathrm{per}} = 10$). Boxplots of the split-variance estimate $\hat{\sigma}$ against separation δ: at the null ($\delta = 0$) both estimators sit at the true $\sigma = 1$ (grey dashed), but as the clusters separate they inflate toward ≈ 3 — the separation being tested leaks into the global variance estimate, so both folds are split with too much noise. Our studentized test needs no such estimate. Source: `sims/plot_datathin_sigma_box.R`.

Figures

Supplementary figures

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